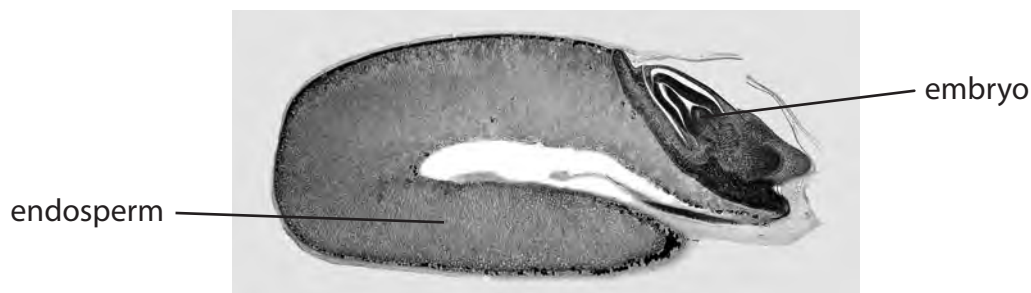


- 4 The photograph shows a section through a wheat grain.



(Source: © DR KEITH WHEELER / SCIENCE PHOTO LIBRARY)

Wheat grains store starch in the endosperm.

During germination the embryo releases amylase from the cells surrounding the endosperm.

The starch is digested by the amylase. This allows the embryo to grow.

Absciscic acid inhibits the germination of wheat grains.

A student formed the following hypothesis:

*the greater the concentration of abscisic acid (ABA), the smaller the production of amylase by the wheat grain.*

Plan an investigation to find evidence to support or reject this hypothesis.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(d) Suggest **two** limitations of your proposed method.

(2)

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(Total for Question 4 = 16 marks)

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**TOTAL FOR PAPER = 50 MARKS**